

PROJECT III

Adapted equipartition calculation

1. Introduction

The diffusive shock acceleration (DSA) theory in the test-particle regime does not account for power-law distributions with energy indices $\Gamma < 2$ (Bell 1978), but assuming that the energy index is not directly related to shock compression, we could try to estimate magnetic field through an adapted equipartition calculation.

The synchrotron flux density can be written as (Arbutina 2017)

$$S_\nu = \frac{4\pi}{3} f \theta^3 d c_5 K_e (B \sin \Theta)^{(\Gamma+1)/2} \left(\frac{\nu}{2c_1} \right)^{(1-\Gamma)/2}, \quad (1)$$

where c_1 and c_5 are defined in Pacholczyk (1970), B is the magnetic field strength, Θ the pitch angle, K_e the constant in the power-law energy distributions of cosmic ray (CR) electrons $N(E) = K_e E^{-\Gamma}$, ν is the frequency, f is the volume filling factor, θ the angular radius and d is the distance. We shall use averaged pitch angle, $\langle (\sin \Theta)^{(\Gamma+1)/2} \rangle = \frac{\sqrt{\pi}}{2} \frac{\Gamma(\frac{\Gamma+5}{4})}{\Gamma(\frac{\Gamma+7}{4})}$.

From the above equation, using spectral index $\alpha = \frac{\Gamma-1}{2}$ we can write

$$K_e B^{\alpha+1} = J, \quad (2)$$

where J is a function that can be calculated from observable quantities. In order to estimate B , we need to know some relation between K_e and B , which is the basis for the minimum-energy or equipartition calculation (Pacholczyk 1970). In supernova remnants (SNRs) equipartition between CRs energy and magnetic field energy does not seem to be reached, but the magnetic field energy density may be comparable to the energy density of CR electrons (Urošević et al. 2018).

CR energy density for a power-law particle distribution of electrons $f(p) = kp^{-\Gamma-2}$ can be written as (Arbutina et al. 2012, 2013, Arbutina 2017)

$$\begin{aligned} \epsilon_{\text{CR}} &= \int_{p_{\text{inj}}}^{p_\infty} 4\pi k p^{-\Gamma} (\sqrt{p^2 c^2 + m_e^2 c^4} - m_e c^2) dp \\ &= 4\pi k c (m_e c)^{2-\Gamma} \int_{\frac{p_{\text{inj}}}{m_e c}}^{\frac{p_\infty}{m_e c}} x^{-\Gamma} (\sqrt{x^2 + 1} - 1) dx, \quad x = \frac{p}{m_e c} \\ &= K_e (m_e c^2)^{2-\Gamma} I_1, \end{aligned} \quad (3)$$

where $K_e = 4\pi k c^{\Gamma-1}$. Injection momentum can be written as $p_{\text{inj}} = \xi p_{th} = \xi \sqrt{2m_e k T_e}$, where ξ is injection parameter related to the injection efficiency (Blasi, Gabici & Vannoni 2005)

$$\eta = \frac{4}{\sqrt{\pi}} \frac{1}{\Gamma - 1} \xi^3 e^{-\xi^2}. \quad (4)$$

For T_e , we can provisionally assume $T_e = \beta T_p$ (Ghavamian et al. 2013), where T_p can be calculated from Rankine-Hugoniot jump conditions

$$kT_p = \frac{(\gamma + 1 + 2\gamma(M_1^2 - 1))(2 + (\gamma - 1)M_1^2)}{\gamma(\gamma + 1)^2 M_1^2} m_p c_s^2, \quad M_1 = \frac{v_s}{c_s} \quad c_s^2 = \gamma \frac{P_0}{\rho_0}.$$

Assuming synchrotron losses, the maximum electron energy is (Zirakashvili & Aharonian 2007)

$$E_\infty = p_\infty c = \frac{1}{1 + \sqrt{\kappa}} \frac{\sigma - 1}{3\sigma} \frac{m_e^2 c^3 v_s}{\sqrt{\frac{2}{27}} e^3 B}, \quad (5)$$

where v_s is the shock velocity, σ shock compression

$$\sigma = \frac{\rho}{\rho_0} = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2},$$

$\kappa = B_0/B = 1/\sqrt{(1 + 2\sigma^2)/3}$, B_0 ambient magnetic field.

If $\epsilon_{\text{CR}} \approx \epsilon_B = \frac{1}{8\pi} B^2$ then

$$K_e = \frac{(m_e c^2)^{\Gamma-2}}{8\pi I_1} B^2, \quad (6)$$

and the (electrons equipartition) magnetic field estimate is

$$B^{\alpha+3} = 8\pi (m_e c)^{2-\Gamma} I_1 J \quad (7)$$

If we had used proton equipartition, then

$$K_p = \frac{(m_p c^2)^{\Gamma-2}}{8\pi I_2} B^2, \quad (8)$$

and with $p_{\text{inj}} = \xi \sqrt{2m_p k T_p}$, $E_\infty = p_\infty c = \frac{3}{8} \frac{v_s}{c} e B_0 R$, $K_{ep} = \left(\frac{m_e T_e}{m_p T_p} \right)^\alpha$ (Arbutina & Zeković 2021), and dimensionless integral I_2 from Eq. (3), but with $x = \frac{p}{m_p c}$, the magnetic field estimate would be

$$B^{\alpha+3} = 8\pi (m_p c)^{2-\Gamma} I_2 J / K_{ep}. \quad (9)$$

2. The problem

Write a Python code that estimates amplified (and ambient) magnetic field for Large Magellanic Cloud supernova remnant J0450-7050 by using:

- a) protons equipartition,
- b) electrons equipartition.

The data are given in a separate file.